Supplementary material for *GridSpike*: defense-aware stress-testing of federated edge intelligence for event classification in IEC 61850 substations

Abstract

This document carries the material that the page budget of the main manuscript could not hold: the formal notation and the equations the body summarizes (Section S1), the extended related-work discussion and the positioning table (Section S2), the full benchmark-realism parameter audit and the generated-federation inventory (Section S3), the battery, provenance and pre-registration detail (Section S4), and the complete set of secondary results tables, the full statistical battery and the auxiliary figures (Section S5). Nothing in it revises a number in the manuscript; every table is generated by the same analysis code from the same merged run corpus.

S1. Notation and formal definitions

This section states the notation the manuscript compresses into prose.

S1.1. Cross-silo federated learning

Let K substations hold private datasets $\mathcal{D}_1, \ldots, \mathcal{D}_K$ with $n_k = |\mathcal{D}_k|$ and $n = \sum_k n_k$. Federated learning minimizes the weighted empirical risk

$$\begin{aligned} \min_w \; F(w) \;&=\; \sum_{k=1}^{K} \frac{n_k}{n} F_k(w), \\ F_k(w) \;&=\; \frac{1}{n_k} \sum_{(x,y)\in\mathcal{D}_k} \ell(f_w(x), y)\,, \end{aligned} \tag{S1}$$

without moving $\mathcal{D}_k$. In FedAvg [1] the server broadcasts $w^{(t)}$, each client runs E local epochs to obtain $w_k^{(t+1)}$ and the server aggregates $w^{(t+1)} = \sum_k (n_k/n)\, w_k^{(t+1)}$. FedProx [2] adds a proximal term to the *client* objective,

$$\tilde{F}_k(w) \;=\; F_k(w) \;+\; \frac{\mu}{2}\big\|w - w^{(t)}\big\|_2^2, \tag{S2}$$

which limits client drift under heterogeneity and reduces to FedAvg at $\mu = 0$. Throughout, $\Delta_k = w_k - w^{(t)}$ is the *update direction* of client k in a round: clients exchange full parameter vectors, but every defense operates on Δ_k, because full vectors are dominated by the shared global component and are nearly collinear.

S1.2. Spiking neurons, coding and SynOps accounting

Neuron model. The spiking classifier uses hard-reset leaky integrate-and-fire (LIF) neurons unrolled over T internal timesteps. For a layer of n neurons with input current I_t,

$$\begin{aligned} &u_t = \beta \odot u_{t-1} + I_t, \qquad s_t = H(u_t - \vartheta)\,, \\ &u_t \leftarrow u_t \odot (1 - \mathrm{sg}[s_t])\,, \end{aligned} \tag{S3}$$

where H is the Heaviside step, $\mathrm{sg}[\cdot]$ is the stop-gradient operator (the reset carries no gradient), and both the decay $\beta \in (0,1)^n$ and the threshold $\vartheta \in \mathbb{R}^n$ are learned per neuron. The non-differentiable step is trained with the arctangent surrogate gradient

$$\frac{\partial \tilde{s}}{\partial v} \;=\; \frac{\alpha/2}{1+\left(\frac{\pi\alpha v}{2}\right)^2}, \qquad v = u_t - \vartheta,\ \alpha = 2, \tag{S4}$$

the documented default recipe of the reference toolkit [3, 4]. Logits are rate-decoded: the linear readout is accumulated over the unroll and divided by T.

Input coding. Two encodings are implemented and ablated [5]. *Rate coding* presents the standardized window at every unroll step as Bernoulli spikes with probability $\sigma(x)$; *temporal-contrast* (delta) coding emits ON/OFF events wherever the first difference along the native time axis exceeds $\pm\theta_\delta$, folding that axis into the T unroll steps. Delta coding is blind to constant offsets by construction – a property, not a defect, for a detector of transitions.

Energy accounting. A spiking layer performs one *accumulate* per synaptic event, whereas the dense reference performs one *multiply–accumulate* per synapse. Counting synaptic operations

$$\mathrm{SynOps} = \sum_{t=1}^{T} \sum_{\ell} \; n_{\ell,t}^{\mathrm{spk}} \cdot \mathrm{fanout}(\ell), \tag{S5}$$

where $n_{\ell,t}^{\mathrm{spk}}$ is the number of spikes presented to synaptic layer ℓ at unroll step t, the estimated energy per inference of the spiking model and of the dense reference are

$$E_{\mathrm{SNN}} = \mathrm{SynOps} \cdot E_{\mathrm{AC}}, \quad E_{\mathrm{ANN}} = \mathrm{MACs} \cdot E_{\mathrm{MAC}}, \tag{S6}$$

and the reported ratio is $\rho = E_{\mathrm{SNN}}/E_{\mathrm{ANN}}$, with the community-standard 45 nm CMOS constants of [6]: $E_{\mathrm{MAC}} = 4.6\,\mathrm{pJ}$ and $E_{\mathrm{AC}} = 0.9\,\mathrm{pJ}$ at 32-bit floating point, and 0.2 pJ and 0.03 pJ respectively for 8-bit integer arithmetic. The constants cover *arithmetic only* – memory access and peripheral energy are excluded – so the result is an estimate via SynOps accounting, not a hardware power measurement.

S1.3. Byzantine-robust aggregation

Krum. [7] scores each update by the sum of squared distances to its $n-f-2$ nearest neighbours,

$$\mathrm{KR}(i) \;=\; \sum_{j\in\mathcal{N}_i} \|\Delta_i - \Delta_j\|_2^2, \qquad |\mathcal{N}_i| = n - f - 2, \tag{S7}$$

and *selects* $\arg\min_i \mathrm{KR}(i)$; multi-Krum averages the m lowest-scoring updates. The guarantee requires $n \geq 2f + 3$.

Geometric median. [8] minimizes $\sum_i \|y - \Delta_i\|_2$ and has breakdown point 1/2. It is computed by the Weiszfeld iteration

$$y^{(m+1)} \;=\; \Big(\textstyle\sum_i \|y^{(m)} - \Delta_i\|^{-1}\Delta_i\Big) \Big/ \Big(\textstyle\sum_i \|y^{(m)} - \Delta_i\|^{-1}\Big), \tag{S8}$$

started at the coordinate-wise median and stabilized with the Vardi–Zhang modification [9] for the case where the iterate coincides with a data point.

S1.4. PBFT message accounting

Agreement among M validators tolerating f faults follows normal-case PBFT [10]: PRE-PREPARE (primary to backups), PREPARE (each backup to all others) and COMMIT (each prepared replica to all others). A replica

is *prepared* once it holds the pre-prepare and $2f$ matching prepares, and *committed* on $2f+1$ matching commits; both quorum conditions reduce to

$$M \;\geq\; 3f+1. \tag{S9}$$

The accounting counts the primary's PRE-PREPARE as its prepare vote, so the quadratic prepare term is $(M-1)^2$ rather than the looser textbook $M(M-1)$, and the message count of one committing round is exactly

$$(M-1) \;+\; (M-1)^2 \;+\; M(M-1) \;=\; 2M(M-1), \tag{S10}$$

so at $M = 10$ a committing round costs exactly 180 messages. A stalled round ($M < 3f+1$) commits nothing – no honest replica becomes prepared and only the f faulty replicas emit commits – and still burns $(M-1)(M+f)$ messages: agreement fails closed, and a round that cannot commit admits no update at all.

The fault model is stated exactly as the simulation realizes it, because it is weaker than the word "Byzantine" suggests: the f faulty validators stay online and keep sending – counted in every message total, the honest case for overhead – but their votes are discarded uniformly, and they vote *against* a deterministic ballot rather than *equivocating*, which a ballot computed from a public score vector cannot express – that would need per-recipient message content, and the $M \geq 3f+1$ safety margin is the same either way. A round completes only when *every* honest validator holds the decision, so the slowest honest replica sets the critical path. The committee's primary (replica 0) is always honest, so no view change is ever triggered; batching, checkpointing, garbage collection and message-authentication cost are not modelled; and

only network time enters the latency. Every one of these omissions makes the reported consensus overhead a **lower bound**.

S1.5. Local differential privacy and Rényi composition

Honest clients may sanitize their own update before submission: the Gaussian mechanism with per-client ℓ_2 clipping [11] scales the update by $\min(1, C/\|\Delta_k\|_2)$ and perturbs it with $\mathcal{N}(0, \sigma^2 C^2 \mathbf{I})$, so σ is a noise *multiplier* relative to the clipping bound C. The declared adjacency is *replace-one*, whose ℓ_2 sensitivity is $2C$: every ε in the paper is about a factor of four worse than the add/remove-one reading, which strengthens rather than weakens the privacy-versus-admission incompatibility reported for H4. Privacy is accounted in Rényi differential privacy [12]: each release is $(\alpha, \alpha/(2\sigma^2))$-RDP, T adaptive rounds compose additively, and the optimal conversion to (ε, δ)-DP is available in closed form,

$$\varepsilon = c + 2\sqrt{c\,L}, \quad c = \tfrac{T}{2\sigma^2}, \quad L = \log(1/\delta). \tag{S11}$$

Full participation is assumed every round, so no subsampling amplification is claimed; the accountant is validated against the classical calibration of [13].

S2. Extended related work and positioning

GridSpike sits at the intersection of five active lines. Each gets a paragraph, with the single most dangerous neighbour for the claims named; Table S1 positions the system explicitly.

Federated learning for power systems. Federated learning for grid fault and disturbance discrimination is established: FeDiSa [14] separates physical

faults from cyberattacks across grid zones, FedDiSC [15] adds DP-SignSGD gradient privacy, clustered federated learning detects false-data injection under heterogeneity [16], surveys map the wider space [17], and federated attack detection has reached cyber-physical protection schemes [18]. All consume fixed datasets; none ships a physics-based simulation that *constructs* the federation. The closest neighbour, FeDiSa [14], already owns the sentence "federated learning for power fault discrimination", so the claim is the reproducible generator with sample-accurate labels and two-axis controllable heterogeneity, not the task.

Spiking networks for power systems and edge time series. Spiking classifiers have been applied to power-system disturbance classification on phasor measurements, with the energy argument made by converting multiply–accumulate operations into accumulate-only synaptic operations [19]. This is the most dangerous neighbour for the energy claim: it is centralized, transmission-level, phasor-based, and in a top grid venue. We differ by being federated, on distribution-substation measurement windows, and by budgeting picojoules and milliseconds against a named protocol envelope rather than reporting a ratio in isolation. Beyond the grid, evolved spiking networks classify multivariate time series at the milliwatt scale on neuromorphic hardware [20], and neuromorphic ideas are explored for microgrid coordination and communication-free protection [21, 22] – control and protection schemes, not federated classifiers. The unrelated *spiking neural P systems* used for grid fault diagnosis are symbolic fuzzy-reasoning membrane systems, not gradient-trained spiking networks.

On encoding we follow the canonical taxonomy [5]: rate coding is noise-

robust but spike-hungry, while temporal-contrast (delta) coding suits slowly varying signals with salient transients – which is what a window of RMS phasor magnitudes around a sag or a fault is. Both are implemented and ablated. Against these spiking designs the reference frame sets a *selective state-space model* (Mamba [23], succeeding [24]), a dense sequence model with input-dependent state transitions and linear-time recurrence: it keeps the spiking energy argument honest, since a spiking classifier is only worth its accuracy cost if a strong dense sequence model does not simply dominate it.

Federated spiking learning. Federated learning with spiking networks was systematized in [25] after the online federated neuromorphic rule of [26]; hierarchical variants report energy reductions [27] and label-skew-aware algorithms address non-IID settings [28]. The closest work combines federated spiking learning with compression under Byzantine attack [29], but is restricted to *non-omniscient* attackers, uses image benchmarks, and has no consensus layer, no differential privacy and no application envelope. We adopt this lineage: the claim is the specialization to substation event classification with a defense-aware adversary and a measured energy/latency budget.

Byzantine-robust and consensus-backed FL. The robust-aggregation canon is settled: Krum and multi-Krum select the update sitting deepest in the honest cloud [7], and the geometric median averages with a one-half breakdown point [8]. Trust bootstrapping against a server-held root dataset is strong when such a dataset exists [30]; in a multi-utility federation it does not, so we score against the coordinate-wise median of the clients themselves and exclude that family with a stated reason. Optimized model-poisoning attacks that

assume knowledge of the defense define the adversary taxonomy we place ourselves in [31, 32], with the cone/inner-product constructions of [33, 34] as the attacker's geometric ancestors.

Combining Byzantine agreement or blockchain with robust aggregation is crowded [35, 36]: neither accounts for the protocol's own overhead, pre-filters with a public score, or faces a defense-aware attacker. We make no novelty claim for the combination – the contributions are the measured overhead of the agreement layer and the honest ablation against the aggregator standing alone.

Differential privacy against Byzantine resilience. That classical differential privacy and Byzantine resilience compose poorly, with the degradation growing in model dimension, is a theorem [37, 38]; engineering both jointly is active work [39, 40]. What no member of this line supplies is a measured operating curve in an application system: for a given noise multiplier and admission threshold, what fraction of honest clients is rejected, what fraction of defense-aware attackers is admitted, and at what privacy budget. That curve is the pillar claimed.

Machine-learning latency against the IEC 61850 envelope. Benchmarking inference latency against the 4 ms fast-message transfer time is established practice – but only for packet-level intrusion detection [41, 42]. This is the most dangerous adjacency, and the distinction is clean: those classifiers consume network packets, ours consume electrical measurement windows; none is quantized, spiking or federated.

The point of the table is not that any single column is unclaimed – several are – but that no prior system occupies the intersection.

Table S1: Positioning against the closest prior systems

System	Domain	Model	Fed.	Non-IID	Byz. agg.	DA-att.	Cons./Ovh.	DP	Energy	Lat.
FeDiSa [14]	ICS/PMU	MLP/AE	✓	(∼)	–	–	–/–	–	–	–
FedDiSC [15]	ICS/PMU	AE	✓	(∼)	–	–	–/–	✓	–	–
Clustered FDIA [16]	Bus sys.	DNN	✓	✓	–	–	–/–	–	–	–
Mahapatra et al. [19]	PMU	SNN	–	–	–	–	–/–	–	✓	–
FedSNN [25]	Img	SNN	✓	✓	–	–	–/–	–	✓	–
HFedSNN [27]	Img	SNN	✓	(∼)	–	–	–/–	–	✓	–
FedLEC [28]	Img	SNN	✓	✓	–	–	–/–	–	–	–
Nguyen et al. [29]	Img	SNN	✓	✓	✓	–	–/–	–	(∼)	–
FLTrust [30]	Img/tab	DNN	✓	✓	✓	✓	–/–	–	–	–
KFC [35]	Img	DNN	✓	(∼)	✓	–	✓/–	–	–	–
BRFL [36]	Img	DNN	✓	(∼)	✓	–	✓/–	–	–	–
Guerraoui et al. [37]	Theory	–	✓	–	✓	–	–/–	✓	–	–
Moore [42]	GOOSE pkt	RNN/AE	–	–	–	–	–/–	–	–	✓
GridSpike (this work)	**Sub**	**SNN + CNN-LSTM**	✓	✓	✓	✓	✓/✓	✓	✓	✓

Domain: PMU phasor measurement; ICS industrial control data; Img image benchmarks; Sub substation measurement windows. *DA-att.*: defense-aware attacker. *Cons./Ovh.*: agreement or ledger layer / its overhead reported. *Energy*: per-inference energy accounting. *Lat.*: latency against a named protocol budget. ✓ present, – absent, (∼) partial.

S3. Benchmark realism detail

No public dataset provides per-substation labelled fault and disturbance waveforms suitable for a K-client federation; that gap is why the benchmark generator exists. Rather than assert realism, every generator parameter is held to a rule: it carries a verifiable source – a normative standard, a public dataset, or a peer-reviewed paper – and is labelled by the *kind* of evidence behind it. Four labels run through Tables S2–S5:

measured-data real or measured-model data used directly (vendored test-feeder models and their published load profiles);

standard a value taken from a normative standard;

fitted-synthetic a synthetic process *fitted to* cited published statistics, never presented as measured data;

modeling-choice a documented engineering approximation with its anchor.

A test asserts that every parameter row has a non-empty source. The labels are not decoration: readers should discount **fitted-synthetic** rows, and the main manuscript's limitations section lists what they cost.

S3.1. Feeders, solver and metering point

Five unmodified distribution test feeders are vendored: the IEEE 13-, 34-, 37- and 123-bus feeders and the IEEE PES European Low Voltage Test Feeder [43], from the DSS-Extensions test-case distribution under a BSD-3-Clause licence. Clients are assigned feeders cyclically, so feeder identity is a feature-skew axis before any label skew. The IEEE 37-bus feeder is an

ungrounded delta system, retained deliberately: on it, ground faults appear as neutral displacement – unfaulted phases rising toward $\sqrt{3}$ per unit, large zero-sequence voltage and negligible zero-sequence current – rather than as fault current. In a federation containing it, the *same label means a different signal shape* at different clients – exactly the heterogeneity a utility consortium faces.

The solver is the DSS-Extensions alternative OpenDSS engine accessed through `OpenDSSDirect.py`, cross-platform as the Linux hosts require; results come from the alternative engine, not the EPRI-supported build, and its version is in every generation manifest. Measurement is at the substation bay: a voltage transformer on the feeder-head busbar and a current transformer on the first line section, mirroring merging-unit placement. The solve is quasi-static at a **60 s** step in daily mode, with regulator and capacitor control resolved through the time-based control queue at the feeders' own delays; loads are constant-power above 0.7 pu and constant-impedance below, keeping deep-fault solves physical and convergent.

Non-convergence policy. A step that fails the power flow is re-solved with raised iteration caps and then with Newton's method; if it still fails, the sample *holds the last valid value* – the repeat-last stale-data convention of substation telemetry front ends, not a numerical convenience – and is counted. A day is aborted only past 5% of its 1440 steps or ten consecutive held steps. Across the battery federations the worst per-federation held rate was 0.016% and the aggregate 0.0059% (358 of 6.05 million sampled steps), concentrated in a few ill-conditioned feeder-days (Table S7).

S3.2. Load, distributed generation and events

Residential demand uses the 100 measured-model one-minute household profiles distributed with the European LV feeder, from the CREST high-resolution demand model [44]. Commercial load is a fitted-synthetic profile anchored to the DOE/NREL medium-office reference building [45]; seasonal structure is fitted to published domestic load-profile classes [46], and per-phase allocation is bounded inside EN 50160 and ANSI C84.1 unbalance limits.

Rooftop photovoltaic generation is sited on 40% of load buses at two penetration levels (15% and 40% of nominal feeder load, alternating across clients), anchored to published hosting-capacity ranges [47], with Haurwitz clear-sky irradiance [48] and a two-state clear/cloudy Markov process fitted to measured minute-scale clear-sky-index and ramp statistics [49, 50]. On high-penetration feeders these cloud dynamics deepen the labelled sag windows [51]; a sag is a source-side per-unit excursion drawn to land in the IEEE 1159 band, with the PV-loss front superimposed, not a purely PV-driven dip.

Events are drawn at a Poisson rate per client-day, the class mix from the client's Dirichlet vector. Fault resistance is sampled from documented distributions – log-uniform 0.05–40 Ω for single line-to-ground faults [52] and 0.01–1.5 Ω for phase-to-phase and three-phase faults [53] – with a floor on the LV feeder, whose ground-fault loop always includes cable and joint impedance. Fault location is drawn proportional to conductor length [54] and discretized to the nearer terminal bus, an OpenDSS constraint stated rather than hidden. Sag and swell magnitudes are targeted so that the *observed* depth at the metering bus, after regulator counteraction, lands in the IEEE 1159

sustained-variation bands [55, 56], and overload is a sustained cold-load-pickup multiplier [54, 57]. Episode durations are per class (Table S4).

No protection equipment is modelled. The generator contains no relay, recloser, fuse or circuit breaker: nothing clears a fault, so an injected fault stays energized for its whole drawn duration, 2–10 minutes, where a real feeder would clear it in cycles to a few seconds. Fault windows are *more* observable here than in the field, so the absolute accuracies of the fault classes are optimistic, and what this benchmark supports is the relative comparison across models and defenses.

The class prior is realistic, and that is the hard part. Events are rare: the generator is tuned so that at least 90% of windows are NORMAL, and the realized share is 94.1% in the reference federation (Table S6) and 93.0–94.1% (mean 93.6%) across the fourteen staged federations. All primary metrics are computed under this prior – macro-F_1 and per-class recall, never accuracy alone. A consequence: the rarest class, FAULT_LG, can have single-digit support in a client's validation split, so its per-class recall is unstable and occasionally zero.

S3.3. Measurement chain and frequency

The classifier sees instrumented telemetry, not clean solver output. The channels are 60 s RMS / synchrophasor-grade gateway telemetry, not IEC 61850-9-2 sampled values at 4–4.8 kHz – no merging unit or sampled-value publisher exists in the generator, so the 61850 framing is substation-automation context, not a point-on-wave claim. Current and voltage channels are perturbed by an accuracy-class-0.5 error model bounded by the class limit

[58, 59]; power channels carry the equivalent combined-chain envelope [60]; frequency error follows the synchrophasor steady-state limit [61]; and all channels are quantized by a 16-bit converter over fixed full-scale ranges, clipping being core saturation [62]. Sensor dropout episodes hold the last value across all channels at measured synchrophasor-network rates [63].

System frequency is *not* a solver output – quasi-static OpenDSS solves fix it – so f_{dev} is a stationary Ornstein–Uhlenbeck process fitted to the open measured-frequency database of [64], with class-dependent event-coupled dips and recoveries: a fitted-synthetic channel, labelled as such everywhere.

S3.4. Federation construction

Windows are $T = 64$ samples at the 60 s cadence – about an hour of acquisition per window – with stride 16, giving $\lfloor (1440 - 64)/16 \rfloor + 1 = 87$ windows per simulated day and 2 610 per client at the 30-day default. Windows never cross a day boundary. Client-day d simulates calendar day $3 + 12d$, so the default spans a full year and cycles all weekday/weekend residues.

Heterogeneity is two-axis. Feeder and load-shape identity gives *feature* skew; a Dirichlet(α) draw over the eight event classes gives *label* skew [65], with $\alpha = 0.5$ as the reference and $\{0.1, 1.0\}$ plus a near-uniform control in the sweep. The draw deliberately overrides the natural fault-type mix: the label skew *is* the federated protocol, not a claim about fault statistics. Splits are per-client 70/15/15 and seeded, and the per-channel affine normalizer is fitted on the training split only. Two protocols exist and both are reported: the pre-registered seeded *random* permutation, and the leakage-controlled *chronological* split – contiguous blocks with a four-window purge at each boundary, the

normalizer refitted on the resulting train block. The distinction is measured, not asserted: the committed script `experiments/measure_leakage.py` finds 99.9% of test windows overlapping at least one training window under the random split, against $\approx 0\%$ under the chronological one.

Inter-validator message latency for the consensus layer is a one-way Log-normal(median 25 ms, $\sigma_{\ln} = 0.5$) truncated to $[5, 200]$ ms, in the form of measured wide-area round-trip-time studies [66] and inside the IEC 61850 wide-area transfer-time class budgets [67]. Experiments always inject this sourced sampler; the library's placeholder distribution backs no reported number.

Scale-up path. The same pipeline runs unmodified on the NREL SMART-DS synthetic distribution systems [68], the declared scale-up reference. The IEEE feeders remain the substrate: universally recognized and small enough for a multi-seed battery.

S4. Battery, provenance and pre-registration detail

S4.1. The eighteen-arm reference frame and the attack grid

Federated results are only interpretable between two bounds, and the surveyed application literature usually reports neither. Both are run: a *centralized ceiling* (pooled data – the best a single model can do if the privacy constraint were lifted) and a *local-only floor* (each substation training alone; mean, minimum and maximum over the ten clients), beside plain FedAvg, FedProx, and the full pipeline with each robust aggregator (Table S8). A federated arm that beats the floor buys something over isolation; the gap to the ceiling is the price of decentralization, privacy and Byzantine tolerance.

Table S2: Simulation realism parameters, part 1: grid, load and distributed generation

Parameter	Value / distribution	Basis	Source
Feeders, solver, metering			
Feeder models	IEEE 13/34/37/123-bus + IEEE PES EU LV, unmodified	measured-data	[43]; DSS-Extensions test cases
Solver	OpenDSS (DSS-Extensions engine), daily mode, time-based control queue	measured-data	DSS-Extensions; EPRI OpenDSS manual
Metering point	VT on feeder-head busbar, CT on first line section	modeling-choice	IEC 61850 merging-unit bay placement
Cadence	60 s quasi-static step	modeling-choice	1-min RMS trend telemetry of substation gateways
Load voltage model	constant-P above 0.7 pu, constant-Z below	modeling-choice	OpenDSS load-model convention
Non-convergence	re-solve, then hold-last; abort past 5% of a day or 10 consecutive	modeling-choice	repeat-last stale-data convention of substation telemetry; no 9-2LE stream is implemented

Part 1 continues in Table S3.

Table S3: Simulation realism parameters, part 1 (continued): load and distributed generation

Parameter	Value / distribution	Basis	Source
Load and distributed generation			
Residential profiles	100 measured-model 1-min daily household profiles	measured-data	CREST demand model [44]
Spot-load aggregation	mean of clip$(P/2, 1, 30)$ distinct profiles	modeling-choice	diversity factors [69]
Commercial profile	night base 0.35, sigmoid ramp, weekend plateau 0.55, AR(1) texture	**fitted-synthetic**	DOE/NREL reference building [45]
Seasonal structure	annual sinusoid, amplitude 0.25, winter peak	**fitted-synthetic**	domestic load profile classes [46]
Per-phase unbalance	$\mathcal{N}(1, 0.05)$ clipped to $\pm 10\%$	modeling-choice	bounded by EN 50160 / ANSI C84.1
PV penetration	15% and 40% of nominal feeder load, alternating	standard/ paper	hosting capacity [47]
Clear-sky irradiance	Haurwitz model	standard/ paper	[48]
Cloud transients	2-state Markov (40/20 min), episode depth $\mathcal{U}(0.15, 0.60)$	**fitted-synthetic**	[49, 50]

Basis **fitted-synthetic**: a synthetic process fitted to the cited statistics, not measured data. Part 2 is Tables S4–S5; the complete table ships with the artifact.

Table S4: Simulation realism parameters, part 2: events

Parameter	Value / distribution	Basis	Source
Events			
Event rate	Poisson per client-day, tuned to $\geq 90\%$ NORMAL windows	modeling-choice	distribution fault statistics [54]
Class mix	categorical from the client's Dirichlet(α) draw	protocol	[65]
LG fault resistance	log-uniform 0.05–40 Ω	paper	[52, 53]
LL/LLG/3PH resistance	log-uniform 0.01–1.5 Ω	paper	[53]
Fault location	$\propto$ conductor length, discretized to nearer bus	modeling-choice	exposure model [54]
LG fault duration	lognormal(med 3 min, σ 0.6), clip $[2, 15]$ min	**fitted-synthetic**	HIF persistence at 1-min cadence [70]
LL/LLG/3PH duration	lognormal(med 2.5 min, σ 0.5), clip $[2, 10]$min	modeling-choice	cadence floor; no field basis – nothing clears the fault
Sag/swell, overload duration	$\mathcal{U}(2, 10)$ min; lognormal(med 15 min), clip $[5, 45]$ min	standard/ paper	IEEE 1159 sustained regime [55]; cold-load pickup [54, 57]
Protection equipment	**none: no relay, recloser, fuse or breaker, so a fault persists for its full duration**	modeling-choice	stated limitation, Section 7
Sag / swell	source-side target so observed depth lands in the IEEE 1159 bands	standard	[55, 56, 51]
Overload	feeder-wide multiplier $\mathcal{U}(1.4, 2.0)$, capped per feeder	paper/ standard	cold-load pickup [54, 57]

Same *Basis* legend as Table S2. Bold row: no protection equipment clears a fault, so fault windows are more observable than in the field and absolute fault-class accuracies are optimistic; read only the relative comparison.

Table S5: Simulation realism parameters, part 2 (continued): measurement chain, frequency and network

Parameter	Value / distribution	Basis	Source
Measurement chain and frequency			
CT / VT accuracy	class 0.5: instrument bias + per-sample noise inside the class limit	standard	[58, 59]
P/Q metering	same 0.5% combined-chain envelope	standard	[60]
Frequency error	$\mathcal{N}(0, 1\,\text{mHz})$ truncated	standard	[61]
ADC	16-bit uniform over fixed full-scale ranges, protective-core current range	standard	[62]
Sensor dropout	Poisson episodes, geometric length, hold-last across channels	**fitted-synthetic**	synchrophasor data quality [63]
Frequency wander	stationary OU, $\sigma = 18\,\text{mHz}$, $\tau = 4\,\text{min}$	**fitted-synthetic**	[64]; continental normal band
Event-coupled f_{dev}	mHz, uniform per event: 3PH -40 to -120; LLG -30 to -100; LG/LL -20 to -80; overload -10 to -50; sag -5 to -20; swell $+10$ to $+50$; exponential recovery	**fitted-synthetic**	ENTSO-E $\pm 50\,\text{mHz}$ normal band; $\sim$100 mHz deterministic-deviation scale [64]
Federation and consensus network			
Non-IID protocol	feeder identity $\times$ Dirichlet(α) label skew	protocol	[65]
Windowing	$T = 64$ @ 60 s, stride 16 $\Rightarrow$ 87 windows/day	contract	derived, Section 3
Splits / normalization	per-client 70/15/15, random or chronological; normalizer fitted on train only	contract	leakage measured, not asserted: `experiments/measure_leakage.py`
PBFT link latency	Lognormal(median 25 ms, $\sigma_{\ln} 0.5$) truncated $[5, 200]$ ms	**fitted-synthetic**	lognormal RTT [66]; [67]

Same *Basis* legend as Table S2.

Table S6: Generated federation at the reference configuration

Client	Feeder	PV (%)	Windows	Train/Val/Test	NORMAL (%)	Classes
0	IEEE 13	15	2 610	1826/391/393	94.44	7
1	IEEE 34	40	2 610	1826/391/393	94.60	8
2	IEEE 37	15	2 610	1826/391/393	93.87	5
3	IEEE 123	40	2 610	1826/391/393	94.87	6
4	EU LV	15	2 610	1826/391/393	94.21	7
5	IEEE 13	40	2 610	1826/391/393	94.37	4
6	IEEE 34	15	2 610	1826/391/393	93.49	7
7	IEEE 37	40	2 610	1826/391/393	94.56	8
8	IEEE 123	15	2 610	1826/391/393	93.22	8
9	EU LV	40	2 610	1826/391/393	93.75	5
All	5 feeders	n.a.	26 100	18260/3910/3930	94.14	8

From the dataset manifests: feeder assignment, window counts, realized class prior and split sizes per client.

Table S7: Solver non-convergence per feeder across the battery federations

Feeder	Feeder-days	Held steps	Held rate (%)
IEEE 13	840	21	0.0017
IEEE 34	840	117	0.0097
IEEE 37	840	185	0.0153
IEEE 123	840	35	0.0029
EU LV	840	0	0.0000
All	4 200	358	0.0059

Held samples repeat the last valid value (repeat-last convention), not a simulated merging unit.

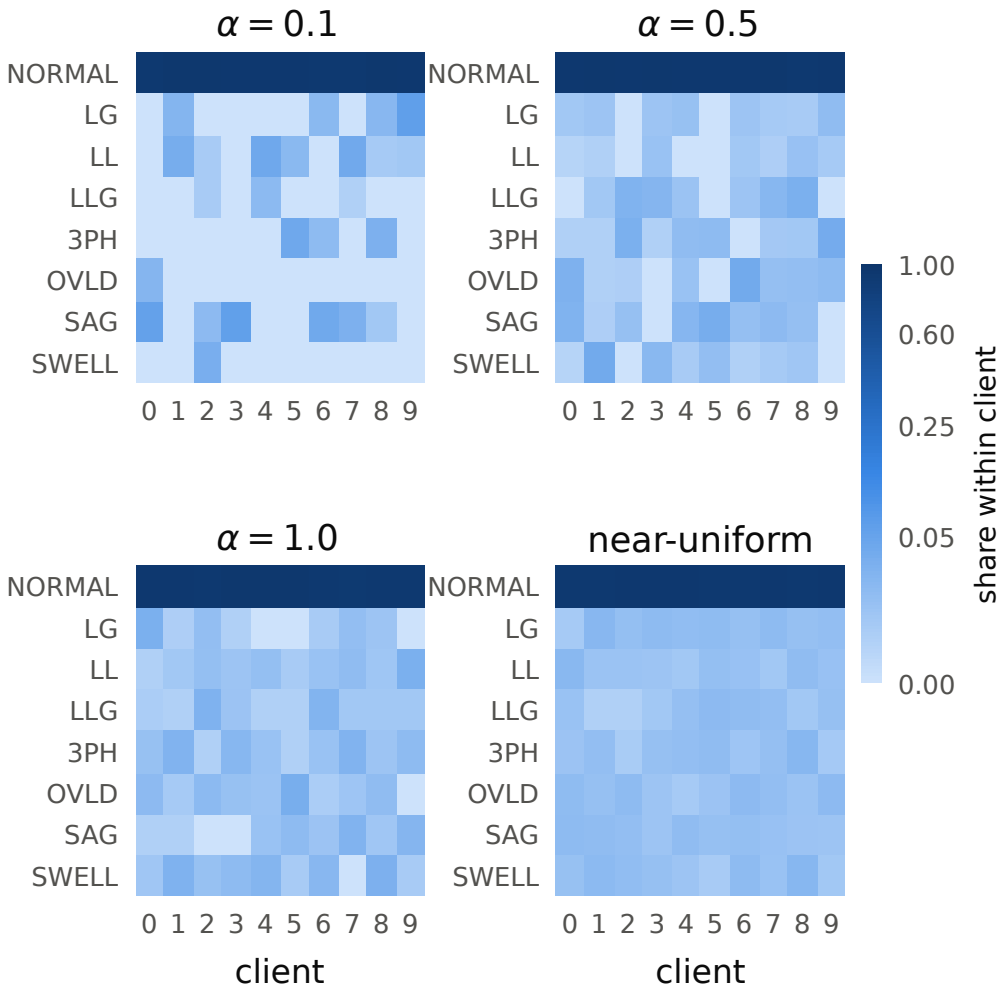


Figure S1: Per-client share of each event class at $\alpha \in \{0.1, 0.5, 1.0\}$ and the near-uniform control, one panel per concentration, ten substations.

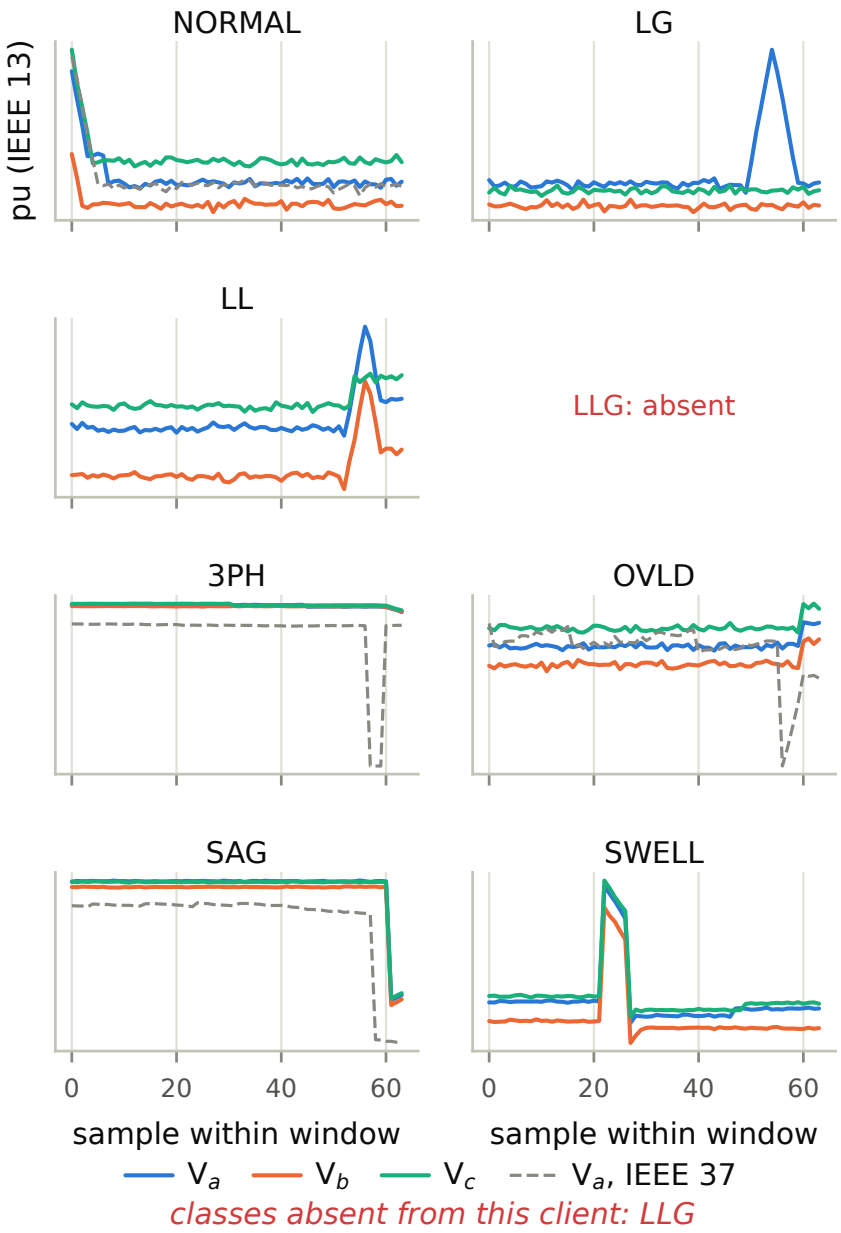


Figure S2: Representative 11×64 windows per event class: phase voltages after the measurement chain, with a cross-feeder reference trace in grey.

Table S8: Reference-frame arms and attack-grid defense arms

Arm	Mode	Aggregation	Role
B1–B3 centralized	pooled	n.a.	accuracy ceiling
B4–B6 local-only	per-client	n.a.	accuracy floor
B7–B9 FedAvg	federated	weighted mean	standard baseline
B10–B12 FedProx	federated	weighted mean (μ term)	heterogeneity baseline
B13–B15 pipeline-Krum	federated	fidelity+PBFT+Krum	proposed
B16–B18 pipeline-geomed	federated	fidelity+PBFT+geo-median	proposed
no-defense	federated	weighted mean	attack grid: what an undefended federation loses
Krum-alone	federated	Krum only	attack grid: does agreement buy accuracy?
geomed-alone	federated	geo-median only	attack grid: same question, averaging aggregator
fidelity-only	federated	gate + weighted mean	ablation: the gate without agreement

Pipeline: fidelity $\rightarrow$ PBFT $\rightarrow$ robust aggregation (Algorithm 1); *alone*: the aggregator with no gate and no agreement layer. n.a.: no aggregation step.

One family is excluded, with a reason rather than silently: root-dataset trust bootstrapping [30] assumes a clean server-held dataset, which is unavailable by design in a multi-utility federation.

S4.2. Battery composition and reconciliation

Table S9 enumerates the eleven battery blocks. Tier 1 – the clean reference frame and the attack × defense grid – carries the headline claims; Tier 2 carries the sensitivity sweeps and ablations. Tier 3 is the *control battery*, specified on 2026-07-31 after the pre-registered analysis was complete: two blocks that re-run the entire Tier-1 grid under a contiguous chronological client split, and one that sweeps the attacker's magnitude. It answers two threats to validity the pre-registered design did not cover – window overlap under a random split, and the one attacker lever a scale-free gate cannot see.

The five run batteries hold 6,836 files between them: the 4,851 distinct seeded runs above, plus six latency benchmarks behind Table S26; the gap between files and runs is re-execution, and deduplication runs in two passes. First by `run_id` – the X3 batteries re-ran the shared seeds (0–4 at Tier 1, 0–2 at Tier 2) under both fresh and inherited identifiers, and on a collision the expansion copy wins (5,535 unique identifiers). Then on a semantic configuration key, because the hash field in the expansion manifests copies its seed line by convention: the T3τ manifest re-emits each (σ, τ) cell once per source T2c threshold (exact triplicates), and three Tier-2e state-space ablations (`ts4`, `ts16`, `codingrate`) are bit-identical configurations measuring the same thing, so its 225 files are 189 distinct runs. Six cells are recorded as *diverged* and excluded from their cell means: the pipeline+Krum state-space LDP runs at $\sigma \in \{1.0, 2.0\}$, $\tau \in \{0.8, 1.0\}$, seeds 0 and 3 (Section 5.9).

Table S9: Battery composition after the X3 extension (merged, deduplicated corpus)

Block	Tier	Grid	Runs
T1a	1	18-arm clean reference frame (6 arms × 3 models)	270
T1b	1	4 attacks × 5 defense arms × 3 models, $f_{\text{att}} = 3/10$, $\alpha = 0.5$	900
T2a	2	$\alpha \in \{0.1, 1.0, \text{near-uniform}\}$ × {FedAvg, pipeline-Krum} × 3 models × {clean, adaptive}	324
T2b	2	$f_{\text{att}} \in \{1, 2\}/10$ × {sign-flip, adaptive} × {no-defense, pipeline-Krum} × 3 models	216
T2c	2	LDP $\sigma \in \{0.25, 0.5, 1, 2\}$ × $\tau \in \{0.3, 0.5, 0.7\}$ × pipeline-Krum × 3 models	324
T2d	2	validators $M \in \{4, 7\}$, $f = \lfloor (M-1)/3 \rfloor$, × 3 models	54
T2e	2	ablations: SNN coding, SNN T steps, FedProx μ, fidelity-only, at 9 seeds	189
T3τ	2	fidelity threshold: 648 attack runs ($\tau \in \{0.6, 0.8, 1.0\}$ × 4 attacks × 3 models × 2 aggregators × 9 seeds) plus 324 local-DP runs (36 distinct $(\sigma, \tau, \text{model})$ cells × 9 seeds, emitted in exact triplicate and deduplicated semantically)	972
Pre-registered subtotal (extended)			**3,249**
T1a_temporal	3	post-registration control: block T1a re-run under the contiguous chronological split	270
T1b_temporal	3	post-registration control: block T1b re-run under the same split	900
T4mag	3	post-registration control: attacker magnitude $s \in \{1, 2, 5, 10\}$ × 4 defense arms × 3 models, adaptive-cosine at $f_{\text{att}} = 3/10$	432
Control subtotal (leakage / magnitude, 2026-07-31)			**1,602**
Total runs carrying a block label			**4,851**

Runs: block inventory of `runs/summary.json`, matching the committed manifests except where noted in the text. Tier 1 and the temporal controls: 15 seeds; Tier 2 and Tier 3 controls: 9.

The control batteries are emitted by the same manifest machinery – each temporal run copies its T1 twin and changes one field, so it pairs with its twin on every other axis.

The sweeps span six sensitivity families – non-IID severity, attack fraction, local-DP noise crossed with the admission threshold, validator committee size, attack type, and arithmetic precision (an analytic re-costing, with no training block of its own) – and eight ablation axes, of which the defense decomposition (none → fidelity-only → Krum-alone → pipeline-Krum → pipeline-geomed) carries the most weight.

S4.3. Seeds, stated openly

Tier 1 and the Tier-3 temporal controls run **15** seeds, and every Tier-2 sweep and the Tier-3 magnitude control run **9**: the X3 coverage extension re-ran the shared seeds (0–4 at Tier 1, 0–2 at Tier 2) and added the new ones, taking the original 5/3 plan – itself reduced by a dated amendment before any Tier-1 run from the 10 and 5 the project specified for itself, on a measured per-run cost – to fifteen and nine. The consequence, plainly: at fifteen paired seeds an exact one-sided Wilcoxon signed-rank can reach 2^{-15}, so every pre-registered family is decidable at $\alpha = 0.05$ after Holm correction (the family floors are 7.3×10^{-4} for H2, 1.8×10^{-4} for H3, 5.9×10^{-3} for H5b), and the verdicts are reads of the data rather than of the test's resolution. One seed drives the *entire* stochastic pipeline – federation generation, initialization, batching, DP noise, attack randomness and the PBFT network draws – so the repetitions are also independent dataset re-draws, which is stronger than k-fold over a frozen split. The extension also mixes two run environments: the original fleet (RTX 3060/3090/4080 class,

torch 2.4–2.13) executed the pre-registered seeds, while the X3 batteries ran on RTX 5090/4090 hardware under torch 2.9.1 – the temporal cells therefore mix environments by construction, and a subset analysis on the homogeneous seeds 5–14 leaves the headline conclusions unchanged.

S4.4. Attack settings

Every attacker is constructed with `collude_seed` equal to the run seed, so all Byzantine clients draw the same ω per (seed, round) – one shared random direction, sign-fixed towards the damaging half-space, that lands the coalition on a single point of the cone; the per-attacker alternative (ω the component of $-h$ orthogonal to u) is implemented but exercised by no run reported here. No manifest entry overrides an attack default, so every attacked run uses target score $\varsigma = 0.1$, safety margin $\eta = 0.15$, magnitude $s = 1$ (swept only in Section 5.8), label-flip rate frac $= 1.0$ and $n_{\text{peers}} = K = 10$, at Byzantine fraction $f_{\text{att}}/K = 0.3$ (0.1 and 0.2 in block T2b).

S4.5. Pre-registration amendments

Five hypotheses were registered before the first Tier-1 run, each with its null, alternative, metric, test, correction family and decision rule (summarized in the manuscript); two supporting claims are reported with numbers rather than tests – the latency envelope and the consensus overhead (a lower bound) – and a third, end-to-end ledger verification, was registered but is *not* instrumented in this battery's result records, so it is reported as a design property with its test coverage named, not as a measured result.

Deviations are recorded as dated amendments with reasons in the released file; four are on record:

- **2026-07-30a**: an analysis-code fix restoring the state-space model to the H2, H3 and H5 test loops, changing no hypothesis, metric or test;
- **2026-07-31a**: the statement that label-flip is realized as dynamic per-batch noise (manuscript threat-model section);
- **2026-07-31b**: the Tier-3 control battery, added *after* the registered analysis was complete – it registers no hypothesis and changes no verdict, and its results are reported as post-hoc controls on the registered ones;
- the **X3 coverage extension** that raised every cell's seed count – it adds runs, not claims: no hypothesis, metric, test or decision rule moved.

S4.6. Provenance audit detail

Every run serializes its complete configuration, its seed, its per-round events, its final metrics, an environment fingerprint (framework and CUDA versions, GPU name) and the git commit that produced it. That last field is the honest weak point: it reads `unknown` in 5,366 of the 5,535 block-labelled records (96.9% – 821 of the 990 records retained from the original battery, and every record of the expansion), because most of the fleet ran from a working tree whose commit was never captured. Per-run code provenance therefore rests on the recorded configuration and environment fingerprint, on the committed manifests and on seeded determinism, not on a commit id – and the configuration hashes corroborate it where they can: 1,231 of the original battery's 1,845 records (66.7%) reproduce their recorded hash under the canonical serializer and the other 614 under an auxiliary generator's

non-canonical JSON encoding, same content, different whitespace – an inconsistency since fixed – so on that battery the audit found no configuration drift. The expansion's manifests copy the seed line's hash by convention, so the canonical audit does not cover them; their identity is the semantic configuration key the corpus is deduplicated on. A same-seed CPU run is byte-identical in its serialized result; bit-reproducibility is claimed *on CPU only*, because CUDA reductions are not deterministic – the original battery's τ-sweep triplicates bound that residual: 24 of its 108 groups differ across replicas at a median non-zero spread near 10^{-4} macro-F_1, maximum 1.2×10^{-3}, while the extension's are byte-identical. Property-based tests assert the invariants the claims rest on – aggregation permutation-invariance and exact weighting, post-clip norm bounds, score range, ledger tamper detection, and that the PBFT simulation commits exactly when $M \geq 3f + 1$ and otherwise fails closed – alongside a golden fixed-seed regression fixture. Training ran on a heterogeneous rented GPU fleet – the original battery on RTX 3060/3090/4080 hardware under torch 2.4–2.13, the extension on RTX 5090/4090 under torch 2.9.1; *all* latency benchmarks ran on a single fixed reference machine while it was otherwise idle. No model was quantized anywhere in this work: the INT8 rows are an analytic re-costing of the same operation counts with 8-bit per-operation constants, and every measured latency is fp32.

S5. Additional results

This section carries every display the manuscript's results section references but could not hold, plus the complete statistical battery. The organi-

zation follows the manuscript.

S5.1. The random-split reference frame and convergence

Table S10 is the pre-registered *random-split* reference frame – the leakage-inflated diagnostic ceiling whose leakage-controlled twin is the headline table of the manuscript. Figure S3 shows the round trajectories of the same battery: only the CNN-LSTM FedAvg and FedProx arms reach 95% of their centralized ceiling inside the 30-round budget, and they do it on every seed.

S5.2. The shallow probe and the frequency-deviation audit

A deep model is only justified if a cheap one cannot do the job, so a shallow gradient-boosted tree was scored on the same five federations – HistGradientBoosting at library defaults, no tuning, over 77 summary-statistic features per window (seven per channel), in fits of 1–4 s. Pooled, it reaches macro-F_1 0.692 ± 0.042 on the same global test set and 0.318 trained per substation; under the chronological cut it falls 23% to 0.533 ± 0.063 – level with the CNN-LSTM centralized ceiling under either protocol and far below Mamba, though the CNN-LSTM's *federated* arm (0.596) still edges the pooled probe under the same split. One disclosure weakens that headline: the 0.692 depends on scikit-learn's default early stopping, which halts the fit at 11 of 100 boosting iterations – disabled, the same probe scores 0.275 ± 0.095. The reading survives both protocols: the reference architecture is not justified by its *centralized* accuracy – its case is timing – and the benchmark demands deep learning only at the level the state-space model reaches.

Table S10: Reference frame, random split (T1a, $\alpha = 0.5$, no attack, 15 seeds)

	Arm	Model	macro-F_1	BCa 95% CI	Accuracy	Rounds to target
B1–B3	Centralized ceiling	CNN-LSTM	0.582 ± 0.100	$[0.535, 0.633]$	0.555	1.7
		Spiking	0.586 ± 0.085	$[0.534, 0.619]$	0.910	20.5
		Mamba	0.849 ± 0.049	$[0.823, 0.871]$	0.973	9.6
B4–B6	Local-only floor	CNN-LSTM	0.292 ± 0.028	$[0.031, 0.493]^{\dagger}$	0.760	n.a.
		Spiking	0.129 ± 0.008	$[0.049, 0.212]^{\dagger}$	0.597	n.a.
		Mamba	0.197 ± 0.013	$[0.106, 0.359]^{\dagger}$	0.760	n.a.
B7–B9	FedAvg	CNN-LSTM	0.600 ± 0.048	$[0.578, 0.625]$	0.955	16.3
		Spiking	0.161 ± 0.021	$[0.151, 0.172]$	0.658	n.r.
		Mamba	0.770 ± 0.037	$[0.750, 0.786]$	0.973	$29.0^{1/15}$
B10–B12	FedProx ($\mu = 0.01$)	CNN-LSTM	0.603 ± 0.056	$[0.579, 0.635]$	0.955	17.0
		Spiking	0.163 ± 0.020	$[0.153, 0.173]$	0.649	n.r.
		Mamba	0.751 ± 0.030	$[0.735, 0.765]$	0.971	$29.5^{2/15}$
B13–B15	Pipeline + Krum	CNN-LSTM	0.476 ± 0.069	$[0.441, 0.510]$	0.896	$20.8^{4/15}$
		Spiking	0.126 ± 0.018	$[0.118, 0.135]$	0.634	n.r.
		Mamba	0.198 ± 0.074	$[0.163, 0.236]$	0.714	n.r.
B16–B18	Pipeline + geomed	CNN-LSTM	0.454 ± 0.067	$[0.423, 0.489]$	0.942	$22.5^{2/15}$
		Spiking	0.136 ± 0.019	$[0.127, 0.145]$	0.651	n.r.
		Mamba	0.156 ± 0.048	$[0.135, 0.181]$	0.738	n.r.

Leakage-inflated diagnostic ceiling; the headline frame is Table S23. Pooled test set, mean over seeds, BCa 95% CI. †Local-only: per-client [min, max] over the ten substations, not a CI. Centralized ceilings use the federated arms' loss and threshold (recipe-matched).

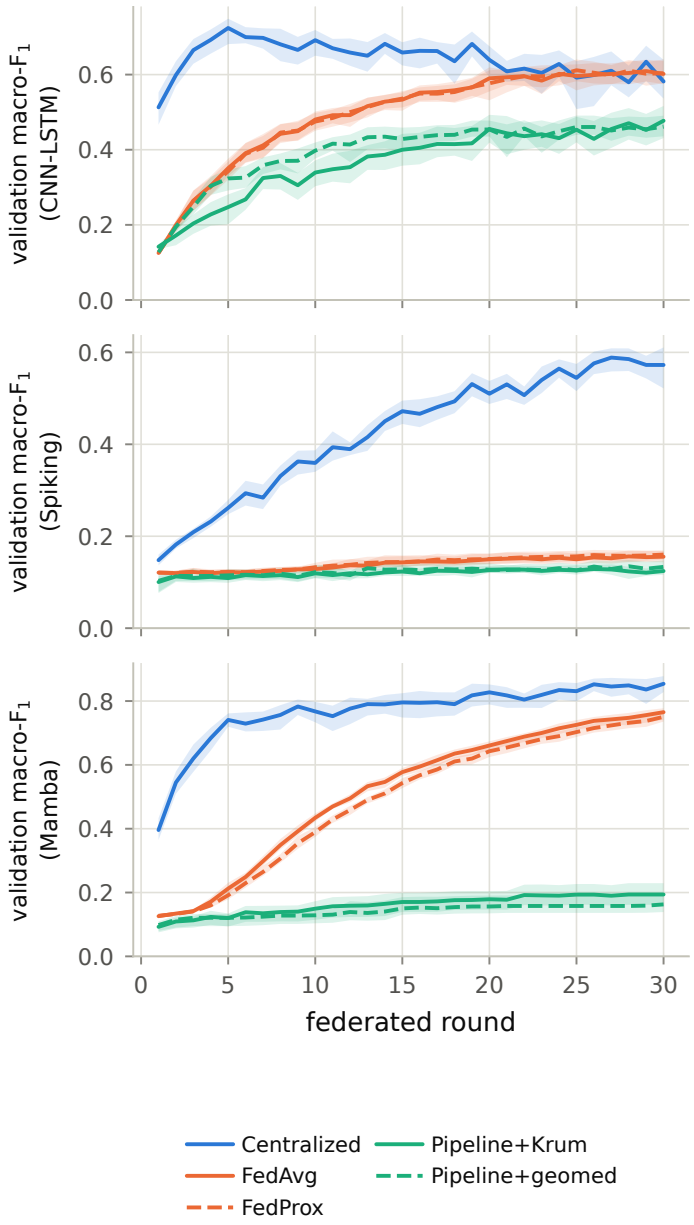


Figure S3: Validation macro-F_1 per federated round (BCa bands), one panel per model (T1a, random split, 15 seeds); trivial-predictor floor macro-F_1 0.121.

Table S11: Frequency-deviation ablation on the shallow probe (5 federations)

Split	Δmacro-F_1	BCa 95% CI	p	f_{dev} only
Random	-0.011	$[-0.061, +0.044]$	0.81	0.295
Temporal	-0.027	$[-0.101, +0.053]$	0.625	0.234

Δmacro-F_1: paired change from dropping the seven f_{dev} features; f_{dev} *only*: refit on those seven alone (majority floor 0.121; all channels 0.692 random, 0.533 temporal). At $n = 5$ the exact two-sided Wilcoxon p cannot fall below 0.0625.

The same probe audits the sharpest shortcut the benchmark offers. The synthetic f_{dev} channel is coupled to event class by construction and could act as a label fingerprint; dropping it costs -0.011 macro-F_1 under the random split and -0.027 under the chronological one, both intervals spanning zero, while the channel alone reaches 0.295 and 0.234 – real but far-from-sufficient signal (Table S11; three limits bound the reading: a shallow probe, not the deep models, whose capacity to exploit a subtle coupling is larger; a test with no power at $n = 5$; and an argument that therefore rests on the magnitude of the change, not its p-value).

S5.3. The random-split attack grid

Table S12 is the attack $\times$ defense grid under the pre-registered random split, at the tolerance-limit attack fraction; its chronological-split counterpart is the headline attack table of the manuscript, printed with the cell-by-cell change against this one. Figure S5 summarizes the multi-arm ranking behind the manuscript's H2 discussion: geometric-median aggregation standing alone attains the best average rank, both full-pipeline arms rank below it, and no-defense is not separated from pipeline+Krum, so the admission pipeline does not improve on robust averaging. The per-seed blocking ($n = 180$)

Table S12: Attack × defense macro-F_1 (mean ± sd), random split (T1b, $f_{att} = 3/10$, $\alpha = 0.5$, 15 seeds)

Defense arm	Model	Clean	Sign-flip	Gaussian	Label-flip†	Adaptive
No defense	CNN-LSTM	0.600 ± 0.048	0.392 ± 0.063	0.059 ± 0.017	0.509 ± 0.069	0.629 ± 0.057
	Spiking	0.161 ± 0.021	0.124 ± 0.012	0.100 ± 0.029	0.093 ± 0.028	0.152 ± 0.018
	Mamba	0.770 ± 0.037	0.293 ± 0.059	0.076 ± 0.010	0.418 ± 0.036	0.739 ± 0.054
Krum alone	CNN-LSTM	n.a.	0.355 ± 0.095	0.408 ± 0.101	0.359 ± 0.074	0.100 ± 0.050
	Spiking	n.a.	0.128 ± 0.020	0.137 ± 0.024	0.124 ± 0.024	0.060 ± 0.062
	Mamba	n.a.	0.224 ± 0.060	0.261 ± 0.042	0.245 ± 0.052	0.043 ± 0.033
Geo-median alone	CNN-LSTM	n.a.	0.437 ± 0.060	0.587 ± 0.043	0.544 ± 0.069	0.609 ± 0.070
	Spiking	n.a.	0.132 ± 0.013	0.151 ± 0.017	0.103 ± 0.025	0.149 ± 0.018
	Mamba	n.a.	0.389 ± 0.071	0.692 ± 0.060	0.439 ± 0.039	0.726 ± 0.037
Pipeline + Krum	CNN-LSTM	0.476 ± 0.069	0.323 ± 0.069	0.366 ± 0.076	0.368 ± 0.065	0.235 ± 0.088
	Spiking	0.126 ± 0.018	0.097 ± 0.039	0.106 ± 0.029	0.095 ± 0.042	0.095 ± 0.036
	Mamba	0.198 ± 0.074	0.076 ± 0.048	0.089 ± 0.057	0.110 ± 0.056	0.031 ± 0.025
Pipeline + geomed	CNN-LSTM	0.454 ± 0.067	0.320 ± 0.051	0.377 ± 0.056	0.357 ± 0.132	0.420 ± 0.140
	Spiking	0.136 ± 0.019	0.097 ± 0.039	0.106 ± 0.028	0.091 ± 0.031	0.082 ± 0.048
	Mamba	0.156 ± 0.048	0.076 ± 0.048	0.084 ± 0.050	0.095 ± 0.049	0.051 ± 0.030

†Realized as dynamic per-batch label noise; see Section 2.5.

separates eight of the ten pairs and preserves this ordering.

S5.4. Defense decomposition

Table S13 decomposes the defense one rung at a time under the defense-aware attacker. Geometric-median aggregation *alone* preserves accuracy almost perfectly while rejecting no honest client, whereas the same aggregator behind the gate and PBFT drops sharply and rejects most honest updates; Krum alone is worse than useless – the single-update selector is exactly what

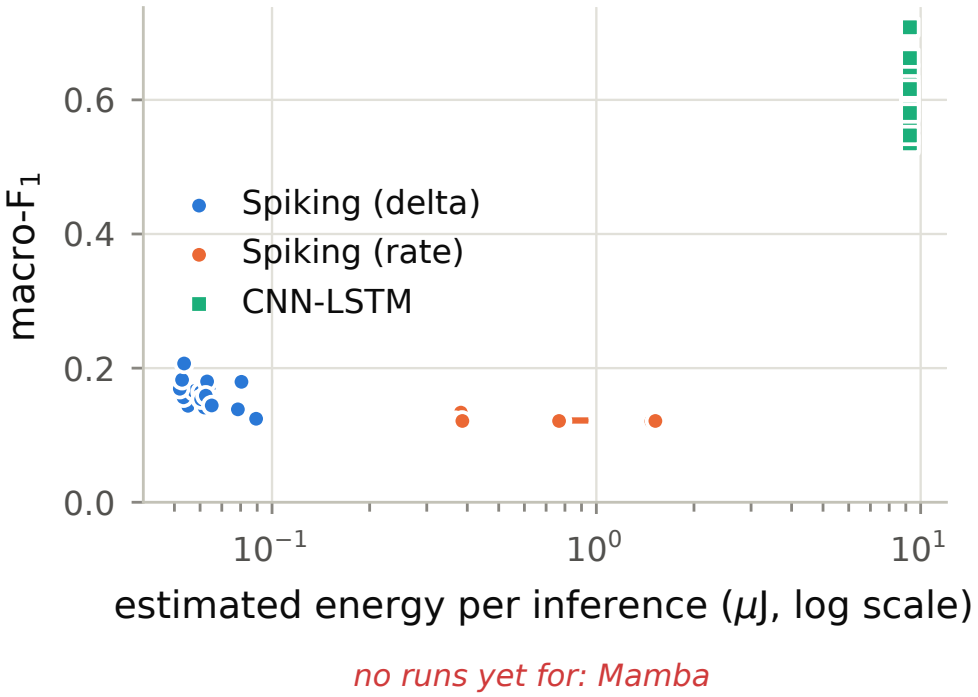


Figure S4: Estimated energy per inference (log scale) against macro-F_1 (BCa intervals) for both spiking codings, CNN-LSTM and Mamba. Energy is an arithmetic-only estimate.

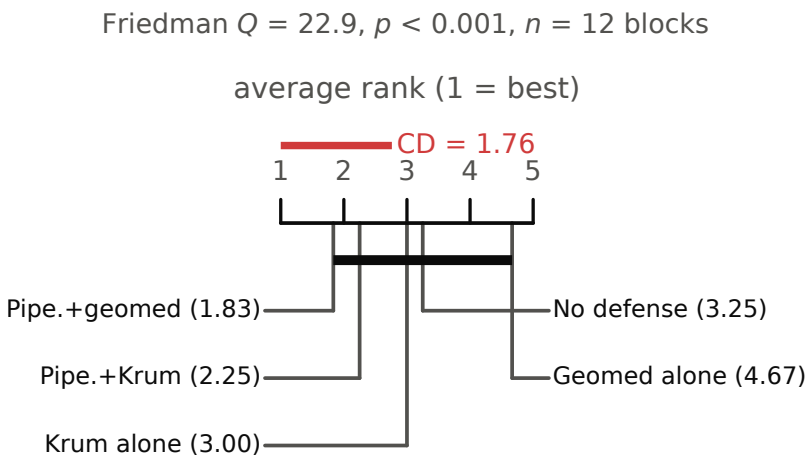


Figure S5: Nemenyi critical-difference diagram over the five defense arms, pre-registered blocking by (attack × model) seed means ($n = 12$, CD = 1.76); bars join indistinguishable arms.

the cosine-aligned attacker games.

S5.5. Defense-aware evasion, in full

S5.6. The admission-threshold sweep

The pre-registered threshold is one point on a curve, and sweeping it changes the diagnosis (Table S15). As τ opens from 0.5 to 1.0 the gate stops rejecting honest clients – false-rejection falls from 0.68–0.97 at $\tau = 0.5$ to 0.11 or less at $\tau = 0.8$ and to 0.016 or less at $\tau = 1.0$ – while the defense-aware attacker is admitted essentially always (admission ≥ 0.92 everywhere at $\tau \geq 0.8$). Behind the geometric median accuracy *recovers* as the gate opens: averaged over the three models, macro-F_1 rises from 0.184 at $\tau = 0.5$ to 0.487 at $\tau = 1.0$, and Mamba – the arm the gate damages most, at 0.051 under the pre-registered calibration – climbs to 0.718, within 0.01 of

Table S13: Defense decomposition, rung by rung, under the defense-aware attacker (random split)

Defense rung	Model	macro-F_1	Δ vs none	Attacker adm.	Honest FRR	Ledger
No defense	CNN-LSTM	0.629 ± 0.057	n.a.	n.a.	n.a.	n.a.
	Spiking	0.152 ± 0.018	n.a.	n.a.	n.a.	n.a.
	Mamba	0.739 ± 0.054	n.a.	n.a.	n.a.	n.a.
Fidelity gate only	CNN-LSTM	0.396 ± 0.110	-0.233	0.606	0.711	n.a.
	Spiking	0.106 ± 0.034	-0.047	0.759	0.744	n.a.
	Mamba	0.064 ± 0.038	-0.675	0.856	0.964	n.a.
Krum alone	CNN-LSTM	0.100 ± 0.050	-0.529	n.a.	n.a.	n.a.
	Spiking	0.060 ± 0.062	-0.093	n.a.	n.a.	n.a.
	Mamba	0.043 ± 0.033	-0.696	n.a.	n.a.	n.a.
Geo-median alone	CNN-LSTM	0.609 ± 0.070	-0.020	n.a.	n.a.	n.a.
	Spiking	0.149 ± 0.018	-0.003	n.a.	n.a.	n.a.
	Mamba	0.726 ± 0.037	-0.013	n.a.	n.a.	n.a.
Fidelity + PBFT + Krum	CNN-LSTM	0.235 ± 0.088	-0.394	0.549	0.691	n.a.
	Spiking	0.095 ± 0.036	-0.057	0.557	0.705	n.a.
	Mamba	0.031 ± 0.025	-0.709	0.846	0.972	n.a.
Fidelity + PBFT + geomed	CNN-LSTM	0.420 ± 0.140	-0.209	0.534	0.657	n.a.
	Spiking	0.082 ± 0.048	-0.070	0.578	0.660	n.a.
	Mamba	0.051 ± 0.030	-0.688	0.846	0.969	n.a.

Admission and honest FRR exist only on gated rungs (fidelity-only and both pipelines); n.a. elsewhere, where they hold by definition. Ledger: n.a. throughout, since no run verifies the hash chain end to end (property tests only).

Table S14: Evasion and damage per attack and aggregator (T1b, random split, 15 seeds)

Aggregator	Attack	Model	Mean score	Admission	Honest FRR	Δmacro-F_1
Krum	Sign-flip	CNN-LSTM	1.046	0.000	0.890	0.153
		Spiking	1.026	0.000	0.958	0.028
		Mamba	0.978	0.000	0.997	0.122
	Adaptive cosine	CNN-LSTM	0.503	0.545	0.708	0.242
		Spiking	0.489	0.568	0.702	0.030
		Mamba	0.407	0.870	0.975	0.167
Geo-median	Sign-flip	CNN-LSTM	1.019	0.000	0.920	0.134
		Spiking	1.015	0.000	0.971	0.039
		Mamba	0.975	0.000	0.997	0.079
	Adaptive cosine	CNN-LSTM	0.552	0.530	0.673	0.034
		Spiking	0.532	0.589	0.655	0.054
		Mamba	0.397	0.870	0.972	0.105

Score: realized fidelity of attacker updates, mean over rounds and seeds; round 1 excluded, since the adaptive attacker then submits its honest direction (Section 2.5).

geo-median-alone (Table S13). Behind Krum it collapses further, 0.120 → 0.069: with a wide-open gate, Krum is again selecting a single update out of a cloud the cosine-aligned attacker was shaped to sit inside. At $\tau = 0.5$ the gate was throttling its own aggregator, discarding most of the honest updates a robust average needs in order to be robust; released, and paired with an averaging-style aggregator, it costs little – $\tau \approx 1.0$ with geometric median is approximately geo-median-alone, with the admission decision still jointly witnessed and written to the chain. The negative result is about calibration rather than mechanism: at the pre-registered operating point the pipeline subtracts accuracy, and no setting of τ makes it *add* any over the bare aggregator.

S5.7. Heterogeneity sweep

S5.8. Consensus overhead

S5.9. The full local-DP operating grid

Table S18 is the pre-registered twelve-point (σ, τ) local-DP grid; its extension to $\tau = 1.0$ is the frontier table of the manuscript.

S5.10. Latency distributions

S5.11. Ablations, in full

Tables S19 and S20 carry the ablation grids. Under delta coding, shortening the unroll helps – $T = 4$ gives the best clean macro-F_1 at 0.204, ahead of $T = 8$ (0.161) and $T = 16$ (0.132), and at nine seeds the ordering is pairwise significant – so longer temporal integration buys more SynOps and

Table S15: Admission-threshold sweep under the defense-aware attacker (random split)

τ	Model	Honest FRR	Attacker adm.	Attacker score	macro-F_1 Krum	macro-F_1 geo-med
0.5	CNN-LSTM	0.691	0.537	0.528	0.235	0.420
	Spiking	0.678	0.579	0.511	0.095	0.082
	Mamba	0.974	0.870	0.402	0.031	0.051
0.6	CNN-LSTM	0.425	0.701	0.491	0.171	0.274
	Spiking	0.438	0.686	0.539	0.091	0.072
	Mamba	0.852	1.000	0.368	0.054	0.031
0.8	CNN-LSTM	0.082	0.925	0.394	0.109	0.584
	Spiking	0.111	0.937	0.310	0.055	0.141
	Mamba	0.064	1.000	0.299	0.056	0.643
1.0	CNN-LSTM	0.000	0.987	0.374	0.061	0.599
	Spiking	0.016	0.969	0.306	0.091	0.145
	Mamba	0.000	1.000	0.287	0.056	0.718

Gate columns pool both pipelines, which share the admission stage. $\tau = 0.5$: T1b (15 seeds); $\tau \in \{0.6, 0.8, 1.0\}$: T3$\tau$ (9 seeds). No local DP here (the $\sigma \times \tau$ grid is Table S18); no diverged runs.

Table S16: Non-IID severity sweep (T2a, random split, 9 seeds)

		CNN-LSTM		Spiking		Mamba	
α	Arm	Clean	Adaptive	Clean	Adaptive	Clean	Adaptive
0.1	FedAvg	0.499	0.532	0.118	0.123	0.798	0.687
	Pipeline + Krum	0.300	0.171	0.113	0.091	0.206	0.051
0.5*	FedAvg	0.600	0.629	0.161	0.152	0.770	0.739
	Pipeline + Krum	0.476	0.235	0.126	0.095	0.198	0.031
1.0	FedAvg	0.649	0.630	0.170	0.171	0.757	0.744
	Pipeline + Krum	0.477	0.233	0.131	0.100	0.180	0.030
near-uniform	FedAvg	0.683	0.629	0.190	0.199	0.770	0.763
	Pipeline + Krum	0.556	0.282	0.139	0.109	0.279	0.050

*Reference-frame value (block T1a).

Near-uniform: a near-uniform Dirichlet draw, not an exact uniform sampler.

Table S17: Consensus cost against committee size M (random split)

			Sim. latency (ms)		macro-F_1 under attack†		
M	f	Messages/round	p50	p95	CNN-LSTM	Spiking	Mamba
4	1	24	113.4	158.6	0.210	0.103	0.023
7	2	84	145.4	199.5	0.220	0.106	0.024
10	3	180	167.8	221.2	0.235	0.095	0.031

†Under the adaptive attacker (30% Byzantine) on pipeline+Krum; not comparable with the clean frame. $M = 10$: T1b (15 seeds); $M \in \{4, 7\}$: T2d (9 seeds). Messages and latency do not depend on the attack. **Lower bound**: always-honest primary, no view change; batching, checkpointing and authentication not modelled.

Table S18: Local-DP operating grid, $\sigma \times \tau$ (T2c, random split, 9 seeds)

					macro-F_1		
τ	σ	ε	Honest FRR	Attacker adm.	CNN-LSTM	Spiking	Mamba
0.3	0.25	345	1.000	0.000	0.024	0.040	0.033
0.3	0.5	113	1.000	0.000	0.024	0.040	0.033
0.3	1.0	41.3	1.000	0.000	0.024	0.040	0.033
0.3	2.0	16.9	1.000	0.000	0.024	0.040	0.033
0.5	0.25	345	1.000	0.000	0.024	0.040	0.033
0.5	0.5	113	1.000	0.000	0.024	0.040	0.033
0.5	1.0	41.3	1.000	0.000	0.024	0.040	0.033
0.5	2.0	16.9	1.000	0.000	0.024	0.040	0.033
0.7	0.25	345	1.000	0.000	0.024	0.040	0.033
0.7	0.5	113	1.000	0.000	0.024	0.040	0.033
0.7	1.0	41.3	1.000	0.000	0.024	0.040	0.033
0.7	2.0	16.9	1.000	0.000	0.024	0.040	0.033

ε: Rényi composition over 30 rounds at $\delta = 10^{-5}$ (Eq. S11); a computed guarantee, not a measurement.

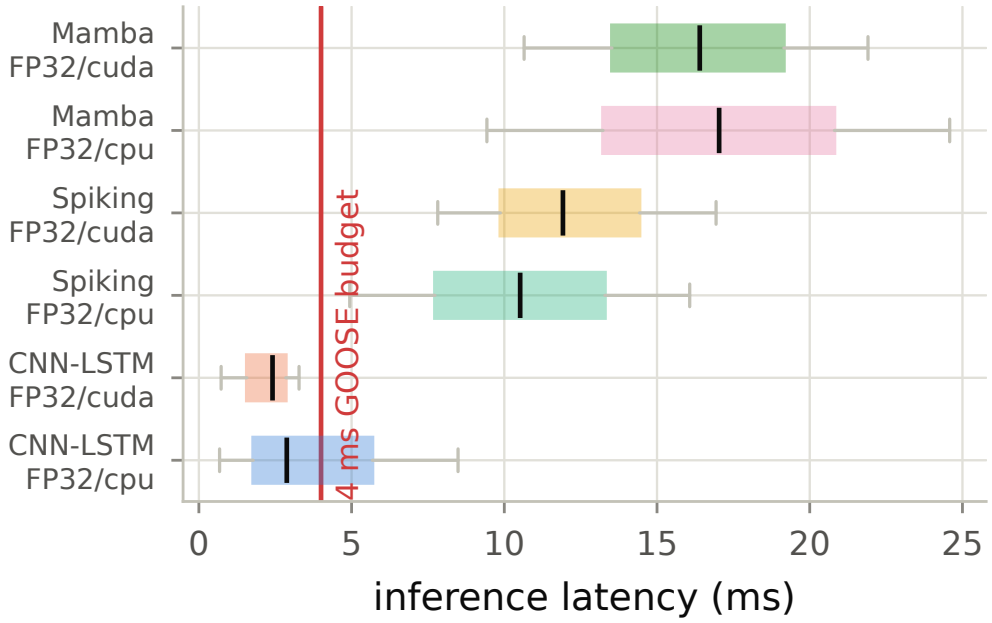


Figure S6: Measured inference-latency distributions per model, precision and device, with p50/p95/p99 marked; vertical line: 4 ms fast-message envelope.

Table S19: Spiking input-coding and unroll-length ablation (T2e, random split, 9 seeds)

Coding	T	macro-F_1 clean	macro-F_1 adaptive	SynOps/inf.	Est. μJ/inf.
rate	4	0.122	0.121	422 827	0.381
rate	8	0.121	0.121	849 521	0.765
rate	16	0.121	0.121	1 693 432	1.52
delta	4	0.204	0.192	47 681	0.0429
delta	8	0.161	0.152	70 079	0.0631
delta	16	0.132	0.121	133 153	0.120

Energy is the arithmetic-only SynOps estimate per inference.

energy for *less* accuracy, and rate coding pins at the 0.121 trivial floor regardless of unroll. FedProx is inert: judged against the *paired* FedAvg baseline (the $\mu \in \{0.001, 0.1\}$ rows run nine seeds against the fifteen-seed reference), no proximal weight moves any model beyond seed noise (largest paired gain +0.006), and the largest weight resolves the harm it always suggested (Mamba at $\mu = 0.1$: 0.614 against a paired 0.764).

S5.12. The complete statistical battery and hypothesis verdicts

S5.13. Exact cells behind the body figures

The body of the paper draws the results below as figures; these tables carry the exact cells – means, standard deviations, realised seed counts and the divergence marks a bar cannot hold – so nothing in the figures is an unverifiable summary. They are the same generated `RunResult` grids the manuscript's previous draft printed in place.

Table S20: FedProx proximal-weight ablation against paired FedAvg (random split)

μ	Model	macro-F_1	BCa 95% CI	FedAvg (paired)	Δ vs FedAvg
FedAvg ($\mu = 0$)	CNN-LSTM	0.600 ± 0.048	$[0.578, 0.625]$	n.a.	n.a.
	Spiking	0.161 ± 0.021	$[0.151, 0.172]$	n.a.	n.a.
	Mamba	0.770 ± 0.037	$[0.750, 0.786]$	n.a.	n.a.
0.001	CNN-LSTM	0.596 ± 0.064	$[0.563, 0.644]$	0.5922^{9}	$+0.004$
	Spiking	0.157 ± 0.022	$[0.146, 0.175]$	0.1575^{9}	-0.000
	Mamba	0.761 ± 0.039	$[0.728, 0.779]$	0.7638^{9}	-0.003
0.01	CNN-LSTM	0.603 ± 0.056	$[0.579, 0.635]$	0.5996	$+0.004$
	Spiking	0.163 ± 0.020	$[0.153, 0.173]$	0.1607	$+0.002$
	Mamba	0.751 ± 0.030	$[0.735, 0.765]$	0.7701	-0.020
0.1	CNN-LSTM	0.586 ± 0.052	$[0.558, 0.623]$	0.5922^{9}	-0.007
	Spiking	0.164 ± 0.019	$[0.154, 0.179]$	0.1575^{9}	$+0.006$
	Mamba	0.614 ± 0.052	$[0.575, 0.640]$	0.7638^{9}	-0.150

$\mu \in \{0, 0.01\}$: T1a (seeds 0–14); $\mu \in \{0.001, 0.1\}$: T2e (seeds 0–8). Δ is the per-seed difference against FedAvg on the shared seeds (*FedAvg (paired)*); a superscript marks fewer than 15 seeds.

Table S21: Complete pre-registered statistical battery (random split)

Family	Comparison	Test	Statistic	p	Holm p	Effect	Band	BCa 95% CI	n
H1a parity	CNN-LSTM vs Spiking, pooled test	McNemar + TOST	16181.58	<0.001	<0.001	−0.439	not equiv.	$[-0.465, -0.417]$	15
H1b energy	ρ vs 1 (one-sided)	BCa CI + Wilcoxon	0.00	<0.001	<0.001	0.007	$\rho < 1$	$[0.006, 0.008]$	15
H2 defense	pipeline+Krum vs no defense (CNN-LSTM/sign-flip) [worst cell]	Wilcoxon + Holm	10.00	0.999	1.000	−0.547	large	$[-0.097, -0.030]$	15
H2 defense	pipeline+Krum vs no defense (CNN-LSTM/Gaussian) [best cell]	Wilcoxon + Holm	120.00	<0.001	0.001	1.000	large	$[0.268, 0.347]$	15
H2 defense	five defense arms, blocks = (attack×model) seed-means [pre-registered]	Friedman + Nemenyi	22.87	<0.001	<0.001	1.761	Nemenyi CD	n.a.	12
H2 defense	same, blocks = per-seed rows [sensitivity]	Friedman + Nemenyi	271.85	<0.001	<0.001	0.455	Nemenyi CD	n.a.	180
H3a evasion	adaptive vs sign-flip score (Krum/CNN-LSTM)	Wilcoxon + Holm	120.00	<0.001	<0.001	1.000	large	$[0.518, 0.565]$	15
H3b damage	adaptive vs sign-flip drop (Krum/Spiking)	Wilcoxon + Holm	52.00	0.524	1.000	−0.044	negligible	$[-0.024, 0.030]$	15
H4 privacy	FRR trend across σ ($\tau = 0.3$)	Jonckheere–Terpstra	0.00	0.500	1.000	0.000	Spearman	n.a.	108
H5a non-IID	macro-F_1 trend across α (FedAvg/Mamba)	Jonckheere–Terpstra	−1.56	0.941	0.941	−0.249	Spearman	n.a.	42
H5b non-IID	FedAvg vs pipeline drop (CNN-LSTM)	Wilcoxon + Holm	9.00	0.951	1.000	−0.506	large	$[-0.126, 0.013]$	9

Holm p: adjusted within family; n: paired seeds. Wilcoxon is one-sided where a direction is pre-registered (H2, H3a, H3b, H5b). Multi-cell families show their least and most favourable cell; 2 of the 24 H2 cells favour the pipeline (Gaussian noise on CNN-LSTM, both aggregators, $\delta = +1.00$).

Table S22: Verdicts against the pre-registered hypotheses

	Pre-registered claim	Evidence	Verdict
H1a	Accuracy parity (equivalence, margin ± 0.02)	McNemar $p < 0.001$, CI $[-0.465, -0.417]$	not supported
H1b	Energy ratio $\rho < 1$ (BCa upper bound)	ρ CI $[0.0063, 0.0076]$	supported
H1	Parity *and* energy advantage	H1a $\wedge$ H1b	not supported
H2	Pipeline beats no defense under each attack	2/24 support (24/24 cells decided)	not supported
H3a	Adaptive attacker evades the fidelity gate	6/6 support (6/6 cells decided)	supported
H3b	Adaptive damage $\geq$ sign-flip damage	2/6 support (6/6 cells decided)	not supported
H3	Evasion *and* damage, at least under geo-median	3/6 support (6/6 geo-median cells decided)	not supported
H4	Honest false-rejection increases with σ	0/3 support (3/3 τ curves decided)	not supported
H5a	Accuracy degrades with heterogeneity	4/6 support (6/6 trends decided)	not supported
H5b	FedAvg degrades more than the pipeline	1/3 support (3/3 models decided)	not supported
H5	Trend *and* differential degradation	H5a $\wedge$ H5b	not supported

Decision rules were fixed before the first run; the full battery is Table S21.

Table S23: Reference frame, chronological against random split (15 seeds)

Arm	Model	macro-F_1 random	macro-F_1 temporal	Δ	Δ (%)
Centralized ceiling	CNN-LSTM	0.582 ± 0.100	0.522 ± 0.104	-0.060	-10.4
	Spiking	0.586 ± 0.085	0.236 ± 0.034	-0.350	-59.7
	Mamba	0.849 ± 0.049	0.724 ± 0.047	-0.125	-14.7
Local-only floor	CNN-LSTM	0.292 ± 0.028	0.270 ± 0.028	-0.022	-7.6
	Spiking	0.129 ± 0.008	0.106 ± 0.006	-0.023	-17.6
	Mamba	0.197 ± 0.013	0.161 ± 0.011	-0.036	-18.2
FedAvg	CNN-LSTM	0.600 ± 0.048	0.596 ± 0.081	-0.004	-0.6
	Spiking	0.161 ± 0.021	0.128 ± 0.020	-0.033	-20.6
	Mamba	0.770 ± 0.037	0.590 ± 0.048	-0.180	-23.4
FedProx ($\mu = 0.01$)	CNN-LSTM	0.603 ± 0.056	0.587 ± 0.073	-0.016	-2.7
	Spiking	0.163 ± 0.020	0.130 ± 0.019	-0.033	-20.4
	Mamba	0.751 ± 0.030	0.585 ± 0.050	-0.165	-22.0
Pipeline + Krum	CNN-LSTM	0.476 ± 0.069	0.401 ± 0.069	-0.075	-15.7
	Spiking	0.126 ± 0.018	0.116 ± 0.015	-0.010	-7.6
	Mamba	0.198 ± 0.074	0.149 ± 0.040	-0.049	-24.9
Pipeline + geomed	CNN-LSTM	0.454 ± 0.067	0.432 ± 0.064	-0.023	-5.0
	Spiking	0.136 ± 0.019	0.117 ± 0.018	-0.018	-13.6
	Mamba	0.156 ± 0.048	0.140 ± 0.033	-0.016	-10.3

Exact cells of Figure 2. Same grid (seeds 0–14, $\alpha = 0.5$, $K = 10$); only the train/validation/test cut differs: uniform (T1a) or chronological (T1a_temporal). Δ: difference of the printed means, unpaired (no shared test set); superscript: seed count below 15. Ceilings are recipe-matched, so the drop is a split effect.

Table S24: Attack × defense macro-F_1, chronological split (T1b_temporal, $f_{\text{att}} = 3/10$, $\alpha = 0.5$, 15 seeds)

Defense arm	Model	Clean	Sign-flip	Gaussian	Label-flip†	Adaptive
No defense	CNN-LSTM	0.596 (−0.004)	0.323 (−0.069)	0.059 (0.000)	0.477 (−0.032)	0.561 (−0.068)
	Spiking	0.128 (−0.033)	0.113 (−0.011)	0.099 (−0.001)	0.074 (−0.019)	0.126 (−0.026)
	Mamba	0.590 (−0.180)	0.227 (−0.066)	0.076 (0.000)	0.326 (−0.092)	0.553 (−0.186)
Krum alone	CNN-LSTM	n.a.	0.348 (−0.007)	0.370 (−0.038)	0.354 (−0.005)	0.093 (−0.007)
	Spiking	n.a.	0.109 (−0.018)	0.111 (−0.025)	0.088 (−0.036)	0.076 (+0.016)
	Mamba	n.a.	0.186 (−0.038)	0.209 (−0.053)	0.220 (−0.024)	0.033 (−0.011)
Geo-median alone	CNN-LSTM	n.a.	0.375 (−0.063)	0.541 (−0.046)	0.515 (−0.029)	0.565 (−0.044)
	Spiking	n.a.	0.114 (−0.018)	0.128 (−0.023)	0.077 (−0.026)	0.126 (−0.023)
	Mamba	n.a.	0.274 (−0.114)	0.512 (−0.179)	0.350 (−0.089)	0.553 (−0.173)
Pipeline + Krum	CNN-LSTM	0.401 (−0.075)	0.324 (+0.001)	0.349 (−0.017)	0.318 (−0.050)	0.207 (−0.028)
	Spiking	0.116 (−0.010)	0.085 (−0.012)	0.095 (−0.011)	0.097 (+0.001)	0.104 (+0.009)
	Mamba	0.149 (−0.049)	0.066 (−0.010)	0.088 (−0.001)	0.108 (−0.002)	0.051 (+0.021)
Pipeline + geomed	CNN-LSTM	0.432 (−0.023)	0.289 (−0.031)	0.349 (−0.028)	0.358 (+0.001)	0.373 (−0.047)
	Spiking	0.117 (−0.018)	0.088 (−0.009)	0.100 (−0.006)	0.096 (+0.005)	0.092 (+0.010)
	Mamba	0.140 (−0.016)	0.061 (−0.015)	0.092 (+0.007)	0.098 (+0.004)	0.074 (+0.023)

Exact cells of Figure 3. Cell: chronological mean; in small type, minus the matched random-split cell of Table S12. †Dynamic per-batch label noise (Section 2.5). n.a.: isolated aggregators run under attack only. Superscript: seed count below 15.

Table S25: Estimated energy per inference

Model	Coding	Precision	SynOps / MACs per inf.	Spike rate	Est. μJ/inf.	Ratio ρ
Spiking	delta (T=8)	fp32	70 079	0.127	0.0631	6.81×10^{-3}
Spiking	rate (T=8)	fp32	849 521	0.366	0.765	0.0825
CNN-LSTM	n.a.	fp32	2 014 208	n.a.	9.27	1.00
Mamba	n.a.	fp32	n.a.	n.a.	n.a.	n.a.
Spiking	delta (T=8)	INT8	70 079	0.127	2.10×10^{-3}	5.22×10^{-3}
Spiking	rate (T=8)	INT8	849 521	0.366	0.0255	0.0633
CNN-LSTM	n.a.	INT8	2 014 208	n.a.	0.403	1.00
Mamba	n.a.	INT8	n.a.	n.a.	n.a.	n.a.

Exact cells of Figure 4. **Arithmetic-only SynOps estimate**, excluding memory and peripherals; not a hardware measurement. Spiking rows at T=8: delta from T1a (15 seeds), rate from T2e (9 seeds). CNN-LSTM MACs are analytic.

Table S26: Measured inference latency per model and device

Model	Device	Mean (ms)	p50 (ms)	p95 (ms)	p99 (ms)	Share of 4 ms env. (%)
CNN-LSTM	CPU	1.108	0.678	2.878	8.486	212.1
CNN-LSTM	GPU	1.013	0.729	2.412	3.280	82.0
Spiking	CPU	5.762	4.940	10.519	16.070	401.8
Spiking	GPU	8.303	7.821	11.920	16.934	423.3
Mamba	CPU	10.552	9.430	17.034	24.582	614.5
Mamba	GPU	11.307	10.649	16.400	21.907	547.7

Exact cells of Figure 5. fp32 only: torch's dynamic INT8 kernels reject the recurrent and state-space layers. One fixed reference machine, 50 warm-up runs; values bound the compute share of the 4 ms envelope, not compliance.

Table S27: Attacker-magnitude sweep (T4mag, random split, 9 seeds)

		macro-F_1			Gate (pooled)		
Scale	Arm	CNN-LSTM	Spiking	Mamba	Atk. score	Atk. adm.	Honest FRR
1×	No defense	0.598 ± 0.064	0.149 ± 0.019	0.727 ± 0.053	n.a.	n.a.	n.a.
	Geo-median alone	0.598 ± 0.076	0.146 ± 0.016	0.719 ± 0.042	n.a.	n.a.	n.a.
	Pipeline + Krum	0.240 ± 0.065	0.099 ± 0.036	0.031 ± 0.029	0.466	0.672	0.799
	Pipeline + geomed	0.282 ± 0.125	0.097 ± 0.038	0.059 ± 0.034	0.495	0.661	0.774
2×	No defense	0.537 ± 0.096	0.147 ± 0.013	0.729 ± 0.050	n.a.	n.a.	n.a.
	Geo-median alone	0.604 ± 0.059	0.151 ± 0.021	0.733 ± 0.058	n.a.	n.a.	n.a.
	Pipeline + Krum	0.145 ± 0.067	0.078 ± 0.056	0.050 ± 0.047	0.491	0.639	0.691
	Pipeline + geomed	0.236 ± 0.114	0.095 ± 0.042	0.053 ± 0.045	0.513	0.636	0.667
5×	No defense	0.138 ± 0.019	0.125 ± 0.008	0.245 ± 0.070	n.a.	n.a.	n.a.
	Geo-median alone	0.598 ± 0.085	0.155 ± 0.021	0.735 ± 0.050	n.a.	n.a.	n.a.
	Pipeline + Krum	0.057 ± 0.049	0.075 ± 0.053	0.049 ± 0.034	0.460	0.651	0.546
	Pipeline + geomed	0.146 ± 0.094	0.117 ± 0.003	0.062 ± 0.033	0.486	0.666	0.570
10×	No defense	0.103 ± 0.025	0.119 ± 0.005	0.205 ± 0.088	n.a.	n.a.	n.a.
	Geo-median alone	0.616 ± 0.054	0.153 ± 0.014	0.738 ± 0.033	n.a.	n.a.	n.a.
	Pipeline + Krum	0.034 ± 0.040	0.083 ± 0.050	0.059 ± 0.035	0.417	0.705	0.476
	Pipeline + geomed	0.060 ± 0.045	0.103 ± 0.040	0.047 ± 0.039	0.446	0.683	0.490

Exact cells of Figure 6; $f_{\text{att}} = 3/10$. *Scale* multiplies the in-cone attacker's update; 1× reproduces the adaptive column of Table S12 at 9 seeds. Gate columns pool the three models, round 1 excluded; n.a. on the two ungated arms.

Table S28: Local-DP grid extended to $\tau = 1.0$ (random split, 9 seeds)

					macro-F_1		
τ	σ	ε	Honest FRR	Attacker adm.	CNN-LSTM	Spiking	Mamba
0.3	0.25	345	1.000	0.000	0.024	0.040	0.033
	0.5	113	1.000	0.000	0.024	0.040	0.033
	1.0	41.3	1.000	0.000	0.024	0.040	0.033
	2.0	16.9	1.000	0.000	0.024	0.040	0.033
0.5	0.25	345	1.000	0.000	0.024	0.040	0.033
	0.5	113	1.000	0.000	0.024	0.040	0.033
	1.0	41.3	1.000	0.000	0.024	0.040	0.033
	2.0	16.9	1.000	0.000	0.024	0.040	0.033
0.6	0.25	345	1.000	0.000	0.024	0.040	0.033
	0.5	113	1.000	0.000	0.024	0.040	0.033
	1.0	41.3	1.000	0.000	0.024	0.040	0.033
	2.0	16.9	1.000	0.000	0.024	0.040	0.033
0.7	0.25	345	1.000	0.000	0.024	0.040	0.033
	0.5	113	1.000	0.000	0.024	0.040	0.033
	1.0	41.3	1.000	0.000	0.024	0.040	0.033
	2.0	16.9	1.000	0.000	0.024	0.040	0.033
0.8	0.25	345	0.967	1.000	0.047	0.029	0.016
	0.5	113	0.967	1.000	0.050	0.025	0.016
	1.0	41.3	0.967^{25}	0.891^{25}	0.050	0.024	$0.088^{7/9,\,2\text{ div.}}$
	2.0	16.9	0.967^{26}	0.788^{26}	0.051	0.023	$0.094^{8/9,\,1\text{ div.}}$
1.0	0.25	345	0.000	1.000	0.047	0.029	0.016
	0.5	113	0.000	1.000	0.050	0.025	0.016
	1.0	41.3	0.000^{25}	1.000^{25}	0.050	0.024	$0.121^{7/9,\,2\text{ div.}}$
	2.0	16.9	0.000^{26}	1.000^{26}	0.052	0.023	$0.121^{8/9,\,1\text{ div.}}$

Exact cells of Figure 7; pipeline+Krum. $\tau \in \{0.3, 0.5, 0.7\}$: T2c; $\tau \in \{0.6, 0.8, 1.0\}$: T3$\tau$ (deduplicated). Identical rows at $\tau \leq 0.7$ are a mechanism: every client is rejected and the model stays untrained. a/b, k div.: converged/planned seeds, k diverged runs excluded. ε as in Table S18.

Table S29: Per-class F_1, recall and support, clean FedAvg (T1a, random split, 15 seeds)

Class	Support	F_1 CNN	Recall CNN	F_1 SNN	Recall SNN
NORMAL	3 674	0.983	0.987	0.800	0.690
FAULT_LG	35	0.511	0.419	0.056	0.147
FAULT_LL	38	0.607	0.524	0.088	0.172
FAULT_LLG	34	0.510	0.413	0.063	0.158
FAULT_3PH	35	0.709	0.648	0.075	0.177
OVERLOAD	36	0.089	0.158	0.051	0.104
SAG	40	0.693	0.564	0.067	0.171
SWELL	38	0.695	0.569	0.084	0.271
Macro	3 930	0.600	0.535	0.161	0.236

Exact cells of Figure 8. Mamba's per-class values are in the run artifacts (its macro-F_1 is in Table S10). Rare fault classes have small support, so their recalls are unstable and reported unsmoothed.

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